Employee Compensation and Workforce Analytics

1. Business Problem Statement

A company wants to better understand its employee distribution, salary structure, payroll expenses, hiring patterns, and organizational hierarchy. The management team needs useful information about how employees are distributed across departments and regions, how salaries differ between jobs, and how much the company spends on employee compensation.

However, raw employee data contains several columns and records that are difficult to understand without proper cleaning and analysis. The company needs a structured approach to transform this data into meaningful business insights.

The project aims to answer the following overarching business question:

“How can the company use employee and salary data to understand workforce distribution, analyze compensation patterns, control payroll expenses, and support better HR and management decisions?”

2. Project Overview

This project analyzes employee information to understand workforce structure, salary distribution, payroll expenses, and employee hiring patterns.

The project uses a sample employee dataset containing 107 employees and 20 original columns. Python is used to clean and prepare the data, MySQL is used to store the cleaned data, SQL is used to answer business questions, and Power BI is used to present the results through an interactive dashboard.

The main goal is to convert raw employee information into meaningful insights that can help management understand the company's workforce and compensation structure.

Project Objectives

- Analyze the number of employees across different departments and regions.
- Calculate the average salary and total payroll expenses.

- Identify the highest-paid employees and jobs with higher average salaries.
- Understand salary distribution and employee salary bands.
- Compare employee salaries with the minimum and maximum salary ranges defined for their jobs.
- Analyze employee hiring trends over time.
- Examine commission patterns and previous job history.
- Understand manager responsibilities and team sizes.
- Compare payroll expenses across different countries.
- Develop an interactive Power BI dashboard to present the findings clearly.

3. Dataset Summary

The project uses a sample HR employee dataset containing employee, job, department, salary, location, and organizational information.

Dataset Details

- **Original dataset:** hr_workforce_dataset.csv
- **Cleaned dataset:** hr_workforce_clean.csv
- **Number of employee records:** 107
- **Number of original columns:** 20
- **Data format:** CSV (Comma-Separated Values)
- **Main tools:** Python, Pandas, MySQL, SQL, and Power BI

Key Features of the Dataset

1. Employee Information

- Employee ID
- Employee name
- Email and phone number, where available
- Hiring date

These fields help identify employees and understand their employment details.

2. Job Information

- Job title
- Job ID
- Minimum salary for the job

- Maximum salary for the job

These fields help compare employee compensation with the defined salary range for each job.

3. Department Information

- Department ID
- Department name
- Manager information

These fields help analyze employee distribution and organizational structure.

4. Salary and Compensation Information

- Employee salary
- Commission percentage

These fields help calculate average salary, total payroll, and commission-related patterns.

5. Location Information

- City
- Country
- Region

These fields help analyze employee distribution and salary expenses across geographical locations.

6. Employment History

- Hiring date
- Previous job history information

These fields help study hiring patterns and employees' previous work experience.

Dataset Preparation

The original dataset is retained separately so that the source information is not overwritten.

Python is used to prepare the data for analysis by handling missing values, converting dates, creating new analytical columns, and removing unnecessary columns from the working dataset.

The cleaned dataset is saved as hr_workforce_clean.csv and is used for the subsequent database analysis.

4. Exploratory Data Analysis Using Python

Python is used to understand, clean, transform, and analyze the employee dataset before loading it into MySQL.

4.1 Data Loading

- Imported the Pandas, Matplotlib, and Seaborn libraries.
- Loaded the original CSV dataset into a Pandas DataFrame.
- Checked the number of rows and columns.
- Examined the available employee information.

4.2 Initial Data Exploration

- Used df.info() to examine column names, data types, and non-missing values.
- Used df.describe() to understand numerical columns such as salary and commission.
- Checked the dataset structure and reviewed the available fields.
- Identified columns that required cleaning or transformation.

4.3 Missing Data Handling

- Checked the dataset for missing values using Pandas.
- Examined missing commission, manager, and location information.
- Replaced missing commission values with zero where appropriate, representing employees without recorded commission.
- Used labels such as Unassigned for missing location information.
- Used No Manager for employees without a recorded manager, where appropriate.

These changes make the dataset easier to analyze while preserving the meaning of the information.

4.4 Date Conversion

- Converted the hiring date into a proper date format.
- Extracted the hiring year into a new column called hire_year.
- Used the hiring year to analyze employee recruitment trends.

4.5 Feature Engineering

Feature engineering means creating new useful columns from existing data.

The project creates the following analytical columns:

1. hire_year

- Extracts the year from the employee's hiring date.
- Helps analyze the number of employees hired in each year.

2. salary_band

- Groups employee salaries into Low, Medium, High, and Very High categories.
- Makes salary distribution easier to understand.

3. has_commission

- Identifies whether an employee receives commission.
- Helps compare employees with and without commission.

4. pay_range_position

- Calculates where an employee's salary falls between the minimum and maximum salary defined for their job.
- Helps understand the employee's position within the job's salary range.

5. pay_level

- Classifies employees according to whether their salary is at or below the job midpoint or above it.
- Helps analyze compensation positioning.

4.6 Data Consistency and Column Selection

- Reviewed the dataset for unnecessary or unsuitable columns.
- Removed columns such as job IDs and contact information from the working dataset when they were not required for the analysis.
- Retained relevant employee, salary, department, location, and employment information.

- Preserved the original dataset separately.

4.7 Exploratory Data Analysis

The project performs several analyses to understand the employee data.

- **Employee count by department:** Identifies how employees are distributed across departments.
- **Average salary by department:** Compares average compensation across departments.
- **Employee count by region:** Examines geographical workforce distribution.
- **Overall salary analysis:** Calculates average salary and total payroll.
- **Salary distribution:** Uses histograms and box plots to understand salary patterns and identify potentially unusual values.
- **Highest-paid employees:** Identifies the five employees with the highest recorded salaries.
- **Hiring trends:** Examines the number of employees hired in each year.
- **Commission analysis:** Compares employees who receive commission with those who do not.
- **Salary range analysis:** Examines employee salary positions within their job salary ranges.
- **Correlation analysis:** Explores relationships between selected numerical variables.

4.8 Data Export

After cleaning and analysis, the prepared dataset is saved as:

hr_workforce_clean.csv

This file is used as the input for the database stage of the project.

5. Data Analysis Using SQL and MySQL

MySQL is used to store the cleaned employee dataset in a structured database. SQL is then used to retrieve, group, filter, and analyze employee information to answer business questions.

5.1 Database Integration

- Connected Python to MySQL using the required database connection libraries.

- Created or used the database named hr_analysis.
- Loaded the cleaned employee dataset into the hr_workforce table.
- Retrieved the data from MySQL to verify the database integration.
- Used the database table for further SQL analysis.

5.2 Business Questions Answered Using SQL

1. Employee Count and Average Salary by Department

- Counted employees in each department.
- Calculated the average salary for each department.
- Compared workforce size and compensation across departments.

2. Employees Earning Above the Overall Average Salary

- Calculated the overall average salary.
- Identified employees whose salaries were higher than the overall average.
- Examined their job titles and salary information.

3. Top Five Job Titles by Average Salary

- Grouped employees by job title.
- Calculated the average salary for each job.
- Sorted the results to identify the five job titles with the highest average salaries.

4. Salary Analysis by Region

- Counted employees in each region.
- Calculated average salary and related compensation measures.
- Compared workforce distribution and salary patterns across regions.

5. Salary Comparison by Commission Status

- Grouped employees according to whether they receive commission.
- Compared employee counts and average salaries.
- Examined differences between the two groups.

6. Hiring Trends by Year

- Grouped employees by hiring year.
- Counted employees hired in each year.
- Identified years with higher hiring activity.

7. Payroll by Salary Band

- Grouped employees into Low, Medium, High, and Very High salary bands.
- Calculated the number of employees and total salary for each band.
- Examined the contribution of each salary group to total payroll.

8. Employees Above the Job Salary Midpoint

- Identified employees whose salaries were above the midpoint of their job salary range.
- Examined their job titles, salaries, and salary-range positions.
- Helped understand compensation positioning without automatically classifying employees as overpaid or underpaid.

9. Managers With the Largest Teams

- Grouped employees by manager.
- Counted the employees reporting to each manager.
- Identified the five managers with the largest recorded teams.

10. Employees With Previous Job History

- Identified employees with one or more previous jobs recorded.
- Examined previous job history information.
- Helped understand the recorded work experience of employees.

11. Payroll by Country

- Grouped employees by country.
- Calculated total salary expenses for each country.
- Compared geographical contributions to total payroll.

5.3 Additional SQL Analysis

The project also supports additional analysis, including:

- Total employee count.
- Overall average salary.
- Total payroll expenses.
- Department-level salary comparisons.
- Highest-paid employee details.
- Regional salary comparisons.
- Commission-related salary patterns.

- Salary-range positioning.

These queries provide structured answers to the business questions and prepare the data for dashboard reporting.

6. Dashboard in Power BI

Power BI is used to convert the analytical results into a clear, interactive dashboard.

The dashboard allows management to understand workforce distribution, salary patterns, hiring trends, and payroll expenses without having to read individual SQL queries.

6.1 Key Performance Indicators (KPIs)

The dashboard includes three main KPI cards.

1. Total Employees

- Displays the total number of employee records.
- Helps management understand the size of the workforce.

2. Average Salary

- Displays the average employee salary.
- Provides a quick overview of compensation levels.

3. Total Payroll

- Displays the sum of employee salaries.
- Helps management understand the total salary cost represented by the dataset.

6.2 Main Dashboard Visuals

1. Employees by Region

- Displays employee distribution across geographical regions.
- Helps management understand where the workforce is concentrated.

2. Average Salary by Department

- Compares average salaries across departments.
- Helps identify differences in departmental compensation.

3. Hires by Year

- Shows the number of employees hired in each year.
- Helps identify periods of higher or lower hiring activity.

4. Employees by Department

- Displays the number of employees in each department.
- Helps management understand departmental workforce distribution.

5. Payroll by Country

- Shows total salary expenses by country.
- Helps compare the payroll contribution of different countries.

6.3 Interactive Slicers

The dashboard includes slicers for filtering the displayed information.

- **Region:** Filters employees by geographical region.
- **Commission Status:** Filters employees with or without commission.
- **Salary Band:** Filters employees by salary category.
- **Department:** Filters the report by department.

When a user selects a slicer value, the connected visuals update according to the selection and the dashboard's configured interactions.

6.4 Benefits of the Dashboard

- Presents important information in a single view.
- Makes salary and workforce comparisons easier.
- Reduces the need to inspect raw employee records.
- Supports interactive exploration of employee information.
- Helps management identify areas that may require further investigation.

7. Key Findings and Business Recommendations

The project produces useful findings about employee distribution, compensation, payroll, and hiring patterns.

7.1 Key Findings

1. Workforce Size

- The dataset contains 107 employee records.
- This provides the basis for analyzing workforce distribution.

2. Total Payroll

- The recorded total salary is 691,400.
- This represents the sum of salaries in the sample dataset.

3. Average Salary

- The recorded average salary is approximately 6,461.68.
- This provides an overall compensation benchmark for the dataset.

4. Regional Distribution

- Americas: 70 employees.
- Europe: 36 employees.
- Unassigned region: 1 employee.

The workforce is primarily concentrated in the Americas in this dataset.

5. Regional Salary Differences

- The recorded average salary for Europe is approximately 8,916.67.
- The recorded average salary for the Americas is approximately 5,191.43.

These differences should be interpreted carefully because employee roles and departmental composition can influence regional averages.

6. Hiring Trends

- The dataset records 28 hires in 1997.
- The dataset records 23 hires in 1998.

These are the two highest hiring-year counts reported in the analysis.

7. Salary-Range Positioning

- The analysis indicates that approximately 77% of employees are below the midpoint of their job salary ranges.

This does not automatically mean that these employees are underpaid. Further analysis of experience, performance, responsibilities, and compensation policies would be required.

7.2 Business Recommendations

1. Review Compensation Structures

- Review salary differences across departments, job titles, and regions.
- Check whether salary ranges are appropriate for different roles.
- Investigate potential inconsistencies before making compensation changes.

2. Improve Payroll Planning

- Monitor total payroll and average salary regularly.
- Compare salary expenses across departments and countries.
- Use the results to support budgeting and workforce planning.

3. Review Salary-Range Positioning

- Examine employees whose salaries are below or above their job midpoint.
- Consider experience, job responsibilities, performance, and company policies before making decisions.
- Avoid treating salary position alone as proof of underpayment or overpayment.

4. Improve Workforce Planning

- Monitor the number of employees in each department.
- Identify departments that may require additional staff or a review of workload distribution.
- Use hiring trends to support future recruitment planning.

5. Review Commission Patterns

- Compare salaries and commission status across relevant jobs.
- Examine whether compensation structures align with job responsibilities and company objectives.
- Use the findings to support a review of commission policies where appropriate.

6. Evaluate Managerial Workload

- Review the number of employees reporting to each manager.
- Identify unusually large teams that may need further workload assessment.

- Consider job complexity and organizational structure before changing team assignments.

7. Improve Data Quality

- Maintain complete and accurate employee, salary, job, and location records.
- Review missing or inconsistent information regularly.
- Use standardized data-cleaning procedures before reporting.

8. Use Power BI for Regular Monitoring

- Refresh the dashboard when updated data becomes available.
- Monitor employee count, average salary, payroll, departmental distribution, and hiring trends.
- Use the dashboard to support recurring management reviews.

8. Project Deliverables

The completed project consists of the following deliverables.

1. Python Data Preparation and Analysis

- Original dataset loading.
- Data-quality checks.
- Missing-value handling.
- Date conversion and feature engineering.
- Exploratory data analysis.
- Charts and summary statistics.
- Cleaned CSV output.

2. MySQL Database

- Database named hr_analysis.
- Structured employee table named hr_workforce.
- Cleaned employee data loaded from Python.

3. SQL Analysis

- Queries for the main business questions.
- Department, salary, region, hiring, and payroll analysis.
- Manager team-size analysis.
- Salary-range and commission analysis.

4. Power BI Dashboard

- Three KPI cards.
- Five main charts.
- Four interactive slicers.
- Visual presentation of workforce and compensation insights.

5. Project Report

- Business problem statement.
- Project objectives.
- Dataset summary.
- Python methodology.
- SQL analysis.
- Power BI dashboard explanation.
- Key findings.
- Business recommendations.

6. GitHub Repository

The project files can be organized in a GitHub repository with folders for Python scripts, SQL queries, dataset documentation, the project report, and Power BI instructions or dashboard files.